

AP CALCULUS AB – UNIT 5

Analytical Applications of Differentiation

AP-Style Review and Exam Practice



Name: _____ Class: _____ Date: _____

Directions

Section I Part A contains 16 non-calculator multiple-choice questions. Section I Part B contains 8 calculator-permitted multiple-choice questions. Section II contains four free-response questions. Show setup, exact reasoning, units, and justification.

Section I – Part A: No Calculator

1. Which hypotheses are required by the Mean Value Theorem on $[a, b]$?

- (A) Continuity on (a, b) only (B) Differentiability on $[a, b]$ only
 (C) Continuity on $[a, b]$ and differentiability on (a, b) (D) $f(a) = f(b)$ and continuity

2. For $f(x) = x^3$ on $[0, 5]$, find a value c guaranteed by the Mean Value Theorem.

- (A) $\frac{5}{3}$ (B) 26
 (C) 5 (D) $\frac{5\sqrt{3}}{3}$

3. If $f'(x) = (x + 2)(x - 1)$, on which interval is f decreasing?

- (A) $(1, \infty)$ (B) $(-2, 1)$
 (C) $(-\infty, -2) \cup (1, \infty)$ (D) $(-\infty, -2)$

4. Suppose f' changes from positive to negative at $x = 3$. What occurs there?

- (A) A relative maximum of f (B) A relative minimum of f
 (C) An inflection point of f (D) A vertical tangent of f

5. If $f''(x) > 0$ on an interval, which statement is true?

- (A) f is concave down and f' is decreasing (B) f is concave up and f' is increasing
 (C) f has a relative minimum everywhere (D) f is increasing and f' is positive

6. Find the inflection point x -coordinate of $f(x) = x^3 - 6x$.

- (A) $\sqrt{2}$ (B) 0
 (C) $-\sqrt{2}$ (D) 3

7. Which set must be checked to find an absolute maximum of a continuous function on a closed interval?

- (A) All points where $f''(x) = 0$ (B) Endpoints and interior critical points
 (C) Only endpoints (D) Only points where $f'(x) = 0$

8. A rectangle has perimeter 58. If its width is x , which area model uses one variable?

- (A) $A(x) = x(58 - x)$ (B) $A(x) = 2x(29 - x)$
 (C) $A(x) = x(29 - x)$ (D) $A(x) = x^2(29 - x)$

9. What dimensions maximize the area of a rectangle with perimeter 32?
- (A) 8 by 8 (B) 16 by 16
(C) 1 by 15 (D) 8 by 9.0
10. At a point where $f'(c) = 0$ and $f''(c) < 0$, the second derivative test concludes
- (A) f has a local maximum at c (B) f has an inflection point at c
(C) no conclusion is possible (D) f has a local minimum at c
11. Rolle's Theorem is a special case of the Mean Value Theorem when
- (A) $a = -b$ (B) $f(a) = f(b)$
(C) $f'(a) = f'(b)$ (D) $f(a) = 0$
12. Which hypotheses are required by the Mean Value Theorem on $[a, b]$?
- (A) Continuity on $[a, b]$ and differentiability on (a, b) (B) Continuity on (a, b) only
(C) $f(a) = f(b)$ and continuity (D) Differentiability on $[a, b]$ only
13. For $f(x) = x^3$ on $[0, 3]$, find a value c guaranteed by the Mean Value Theorem.
- (A) 10 (B) 1
(C) $\sqrt{3}$ (D) 3
14. If $f'(x) = (x + 2)(x - 4)$, on which interval is f decreasing?
- (A) $(-\infty, -2)$ (B) $(-2, 4)$
(C) $(4, \infty)$ (D) $(-\infty, -2) \cup (4, \infty)$
15. Suppose f' changes from positive to negative at $x = 3$. What occurs there?
- (A) A relative minimum of f (B) An inflection point of f
(C) A relative maximum of f (D) A vertical tangent of f
16. If $f''(x) > 0$ on an interval, which statement is true?
- (A) f has a relative minimum everywhere (B) f is concave down and f' is decreasing
(C) f is increasing and f' is positive (D) f is concave up and f' is increasing

Section I – Part B: Calculator Permitted

17. A calculator graph shows $f'(x) > 0$ for $x < -0.6$, $f'(x) < 0$ for $-0.6 < x < 2.4$, and $f'(x) > 0$ for $x > 2.4$. Which statement is true?

- (A) f has an inflection point at each point
(B) f has a relative maximum at $x = -0.6$ and a relative minimum at $x = 2.4$
(C) f has maxima at both points
(D) f has minima at both points

18. A table gives $f(1.8) = 0.1$ and $f(5.3) = 7.5$. What derivative value is guaranteed somewhere between by the MVT, assuming its hypotheses?

- (A) 3.8
(B) 2.114
(C) 3.55
(D) 7.4

19. A cylinder has radius 4.44 and height 7.59. Which value is its volume to three decimals?

- (A) 105.87
(B) 273.673
(C) 211.741
(D) 470.065

20. For $f(x) = x^3 - 3x^2 + 2$, use a calculator to find $f''(0.55)$.

- (A) -1.7
(B) 3.3
(C) -2.393
(D) -2.7

21. A calculator optimization of $A(x) = x(20.90 - x)$ gives which maximum area?

- (A) 109.202
(B) 20.9
(C) 218.405
(D) 54.601

22. A calculator graph shows $f'(x) > 0$ for $x < -0.6$, $f'(x) < 0$ for $-0.6 < x < 2.0$, and $f'(x) > 0$ for $x > 2.0$. Which statement is true?

- (A) f has minima at both points
(B) f has a relative maximum at $x = -0.6$ and a relative minimum at $x = 2.0$
(C) f has an inflection point at each point
(D) f has maxima at both points

23. A table gives $f(1.5) = -0.8$ and $f(5.2) = 6.0$. What derivative value is guaranteed somewhere between by the MVT, assuming its hypotheses?

- (A) 2.6
(B) 6.8
(C) 1.838
(D) 3.35

24. A cylinder has radius 5.81 and height 2.5. Which value is its volume to three decimals?

- (A) 45.632
(B) 265.12
(C) 197.311
(D) 91.263

Section II – Free Response

Questions 1–2: Calculator permitted. Questions 3–4: No calculator.

FRQ1 – Mean Value Theorem

Let $f(x) = x^3 - 3x$ on $[-1, 2]$.

(a) Verify the hypotheses of the MVT. [2 points]

(b) Find the secant slope. [1 points]

(c) Find all c satisfying the conclusion. [2 points]

FRQ2 – First-derivative sign chart

The derivative is $f'(x) = (x + 2)(x - 1)^2(x - 4)$.

(a) Construct a sign chart for f' . [3 points]

(b) State intervals of increase and decrease. [2 points]

(c) Classify each critical point. [3 points]

FRQ3 – Concavity and inflection

Suppose $f''(x) = 6x(x - 3)$.

(a) Determine intervals of concavity. [3 points]

(b) Identify possible inflection inputs and justify. [2 points]

(c) What can be concluded about f' on each interval? [2 points]

FRQ4 – Optimization: open box

A 24 by 18 rectangle has squares of side x cut from each corner and is folded into an open box.

(a) Write the volume and physical domain. [2 points]

(b) Find critical points of the volume. [3 points]

(c) Justify the absolute maximum. [2 points]